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TOPICAL REVIEW

Atomic and molecular processes in the early Universe

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Abstract

Most of the information about the environment of the early Universe comes to us from radiation emitted from atoms and molecules. An understanding of the relevant atomic and molecular processes is needed to correctly interpret this radiation. Atomic and molecular process also control the evolution of the early Universe. In this paper, we review the atomic and molecular processes that are important in the early Universe.

1. Introduction

The standard Big Bang model very successfully accounts for elemental abundance and cosmic background radiation (CBR) measurements in addition to the large scale velocities of galaxies. What is still needed is an understanding of how to get from the relative uniformity, just before the recombination epoch, to the hierarchical structures of clusters, galaxies, and voids that we see in the Universe today. We still do not have a good understanding of how the first objects formed or how galaxies and clusters came to be. To answer these questions we must follow the collapse of initial perturbations in density. A density perturbation that collapses without cooling quickly comes into virial equilibrium. To collapse further requires cooling. Thus the scale of any object formed in the Universe is controlled by cooling, predominantly molecular, and therefore by the gas-phase chemistry of molecule formation and destruction.

In addition to controlling the evolution of these objects, atoms and molecules often provide the best probe of their evolution. Atoms and molecules emit at very specific energies and at rates which are for the most part well known. These then provide the most sensitive measures of the excitation rates and so may, with proper modelling, be used to probe the local gas temperature, density, electron temperature, and radiation field. Additionally, the specific energies may be used to place constraints on any possible change of fundamental physical parameters with time.

Saslaw and Zipoy (1967) were the first to recognize the importance of H_2 cooling on the evolution of gas in the early Universe. They proposed that H_2 was formed through H_2^+,

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the radiative association rate of which had been calculated by Bates (1951), and were able to estimate the cooling thanks to a calculation by Takayanagi and Nishimura (1960). They also considered HeH^+ and H^-, but decided only the H_2^+ route was important. Peebles and Dicke (1968) were the first to realize that formation through H^- was important at later times, following up on McDowell (1961) and Pagel (1959), who had used this reaction pathway to calculate H_2 abundances in the interstellar medium and the Sun respectively—they both reference Dalgarno (1958 private communication) as the source of the inspiration—and the associative detachment rate for H^-. Hirasawa *et al* (1969) independently arrived at H^- as the main source of H_2. A number of groups produced similar models (Takeda *et al* 1969, Hirasawa 1969, Hutchins 1976, Silk 1977, Carlberg 1981). Palla *et al* (1983) were the first to include three-body reactions. Lithium chemistry was first considered by Lepp and Shull (1984) and Dalgarno and Lepp (1987).

There have been reviews of the chemistry of the early Universe by Dalgarno and Lepp (1987), Black (1990), Shapiro (1992), Dalgarno and Fox (1994) and Lepp and Stancil (1998). Still there is much that is not understood. Recently there have been two groups who have assembled comprehensive sets of reactions for the early Universe (Galli and Palla 1998, Stancil *et al* 1998). Most of the rate coefficients are still unknown, and so there is continued work on both calculating these rate coefficients as well as constructing models of collapsing clouds in the early Universe. Peebles (1993) provides a good overview of cosmology and Rees (2000) covers current research in the field.

In this article we review the atomic and molecular processes that are important in the early Universe. It is from the atomic and molecular standpoint a relatively simple system. The Big Bang models produce only three elements and only a few isotopes. Still, considering the few species that need to be modelled, it requires a surprisingly complicated chemistry. We have identified approximately 200 reactions which contribute to the abundance of 23 atomic and molecular species which might be important in the early Universe. Though a smaller reaction set may well reproduce the important chemistry and cooling, it is only by assembling as full a reaction set as possible that one can be certain important reactions are not missing.

We start by reviewing the conditions of the early Universe through the recombination era. Next we consider atoms and molecules as probes to conditions of the early Universe. Then we look at the standard chemistry used to describe the first molecules, including some of the unusual pathways that have been proposed for the early Universe. The cooling processes likely to be important in the collapse of the first objects are covered next, then we look at the collapse of those objects. Finally we consider the effect these first collapsed objects have on their environment.

2. The recombination era

The size of the Universe scales as $R \propto (1 + z)^{-1}$ where R is a characteristic length and z is the redshift. The redshift of an object is a measure of its velocity and in the standard Big Bang cosmology, the velocity of an object is proportional to its distance and therefore the time it takes light to reach us from that object. The temperature of the CBR scales with redshift as $T_r = T_0(1 + z)$, where T_0 is the current CBR temperature. The temperature of the radiation at the present time is approximately 3 K. The current average density of hydrogen atoms in the Universe is of order 10^{-6} cm^{-3}. It scales as the inverse cube of the size, and so the number density at earlier times was $n = n_0(1 + z)^3$. At recombination the density of hydrogen was approximately 10^9 times larger, or of order 10^3 cm^{-3}. This is still a very low-density environment by everyday standards and would be considered an ultrahigh vacuum in the laboratory.

The recombination era took place when the Universe first cooled to a temperature low enough that atoms could survive. At earlier times the temperature of the Universe was hot enough that any atoms formed were immediately reionized by the background radiation. The recombination era began at a redshift of about $z = 2500$ when the Universe was about 100 000 years old, and the temperature was about 4000 K.

The Universe made a transition from an ionized plasma to a mostly neutral gas at recombination. Before the recombination era the Universe was opaque with matter strongly coupled to the radiation via Compton scattering on electrons. After recombination it became transparent and the matter and radiation become decoupled. Cosmic microwave background measurements see back to this opaque wall, the so-called last scattering surface, and measure the properties of the Universe at recombination. The radiation measured by the Cosmic Background Explorer (COBE) originated at that time. The measurements indicate that the radiation at the present time is at a temperature of $T_0 = 2.728$ K (Fixsen *et al* 1996) and that it is highly isotropic with fluctuations only at the 10^{-5} level (Smoot *et al* 1992). Since the radiation and density were highly coupled until recombination we may infer that the Universe at that time was highly uniform with fluctuations in both temperature and density of this same order.

The strong coupling between matter and radiation before recombination also means that before recombination, the temperature of matter and radiation were the same. Before this time, density perturbations were damped out and prevented from collapsing by this coupling to the radiation field.

The recombination era represents the beginning of an epoch where the evolution of the Universe was controlled by atomic and molecular processes. The matter temperature $T_{\rm m}$ decoupled from the radiation temperature $T_{\rm r}$ and density perturbations began to grow. The temperature of the matter was primarily controlled by adiabatic expansion, but atomic and molecular heating and cooling processes also played a role.

In the standard Big Bang model, the hydrogen in the Universe underwent a brief period of nucleosynthesis when the Universe was a few minutes old (Peebles 1966, Wagoner *et al* 1967). The densities were too low to get past the region of instability at mass 8 and so the only elements formed were helium and lithium with a relative abundance of approximately 0.1 and 10^{-10} respectively. Heavier elements are not made until much later when the first stars formed which produced heavy elements in their cores and during the explosive nucleosynthesis of supernovae.

The fractional abundances of primordial atoms and atomic ions are shown in figure 1 for the recombination era. As the radiation temperature of the Universe reduced, atoms and atomic ions appeared in order of their ionization potentials. The ions and atoms and their ionization potentials in order of their appearance are Li^{2+}, Li^{+}, He^{+}, He, H, and Li with ionization potentials of 122.4, 75.6, 54.4, 24.6, 13.6, and 5.4 eV, respectively. Thus the first bound electron was formed when Li^{3+} combined with an electron through the radiative recombination process

$$Li^{3+} + e^{-} \rightarrow Li^{2+} + \nu, \tag{1}$$

to form Li^{2+}. Li^{2+} then recombined to form Li^{+}. Next, He^{2+} and an electron combined to form He^{+}. Later the He^{+} recombined to form He, the first neutral atoms in the Universe. Since hydrogen was still ionized, most of the electrons were still free at the time and these processes continued until completion. At this time, neutral helium atoms can react with ions to form the first molecular ions as discussed below.

After helium, the next to recombine are hydrogen atoms from protons and electrons:

$$H^{+} + e^{-} \rightarrow H + \nu. \tag{2}$$

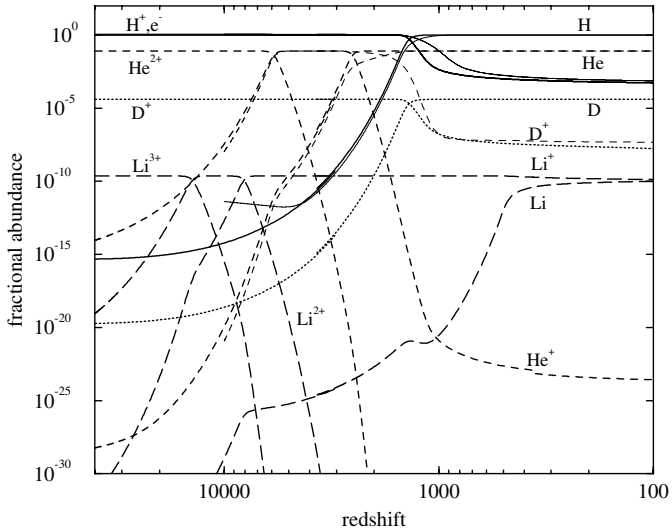


Figure 1. Fractional abundances of neutral and positive atomic ions in the early Universe at high redshift z. The cosmological parameters correspond to the standard cosmology model III of Stancil *et al* (1996, 1998): $\Omega_0 = 1$, $\Omega_b = 0.0367$, $h = 0.67$. This work (thick lines) and Seager *et al* (1999, 2000) (thin lines).

The rate at which this reaction occurred depended on the ionization fraction of the Universe. The actual situation was complicated by the fact that the emitted photon in each of these cases is capable of reionizing the neutral species. A recombination directly to the ground state just generates a photon which reionizes another neutral hydrogen. If we make the approximation that the photon was reabsorbed 'on-the-spot' then this is equivalent to no recombination. For a recombination to lead to a neutral hydrogen it must instead recombine to an excited state and cascade down:

$$\mathrm{H}^+ + \mathrm{e}^- \rightarrow \mathrm{H}^* + \nu_1, \tag{3}$$

$$\mathrm{H}^* \rightarrow \mathrm{H} + \nu_2. \tag{4}$$

This is called case B recombination. Even here, the photon with frequency ν_2 can excite another neutral hydrogen. And with the lower ionization potential, the excited state hydrogen may be reionized by the background radiation. We only gain another ground state hydrogen if the cascade reaction (4) instead occurs via a two-photon transition, or a collisional de-excitation, or the photon ν_2 redshifts out of the window for re-exciting an atom. Detailed models of the recombination of H and He have been made by Seager *et al* (1999, 2000) in which 100-level atoms were considered, including all bound–bound and bound–free transitions. They found that the ionization fraction of the early Universe is about 6% smaller than standard calculations and that the recombination of He was delayed somewhat until just before H recombination (see figure 1).

The recombination of hydrogen is limited because the ionization fractions eventually fall so low that the time to recombine is long compared with the evolution time for (or age of) the Universe. This is because the electron density has fallen to such a low value. This not only slows hydrogen recombination, but all recombinations are slowed. And so Li^+, which has an ionization potential less than that of hydrogen, never fully recombines (Dalgarno and Lepp 1987). It will only completely recombine in regions which collapse, due to the increase in density.

3. Atomic and molecular probes

Astronomers are hampered in their research by their inability to have direct contact with the material with which they are working. Unable to insert measuring devices into the experiment they must instead infer conditions by observed emissions. Atoms and molecules with their emission at specific frequencies provide our best probes into the conditions of the early Universe.

One of the important constraints on the Big Bang model is the abundance of the light elements. A few minutes after the Big Bang the Universe produced helium and smaller amounts of deuterium and lithium (Peebles 1966, Wagoner *et al* 1967). The measurements and their implications for the Big Bang have recently been reviewed (Pagel 2000, Tytler *et al* 2000, Hobbs 2000). The fractional abundances produced are sensitive to the baryon density of the Universe, with larger baryon densities giving more helium and less deuterium; deuterium is the most sensitive and provides the best constraint. The fractional lithium abundance has a minimum near the most likely baryon density and so is not as sensitive.

Ideally one would like to measure the abundances of these elements shortly after they were produced. This is not possible but it is possible to see abundances back to a redshift of about five, thanks to quasars. Quasars are believed to be galaxies with massive black holes at their centre: as the black holes accrete gas they produce large amounts of continuum emission, providing a nearly ideal background source through which to study intergalactic gas. By searching for clouds with relatively low abundances of heavy elements we can get a good measure of the initial elemental abundances. Astronomers use the term metal to refer to any elements heavier than helium; thus these clouds are referred to as metal-free clouds. Observations of relatively metal-free clouds seen in absorption against high-redshift quasars have measured the deuterium abundance. Measurements of this sort are very difficult, and early reports varied widely in their D/H ratios. It now appears that all the measurements are consistent with $\mathrm{D/H} \approx 3 \times 10^{-5}$ (Tytler *et al* 2000, Pettini and Bowen 2001).

In addition to abundances, by measuring excitation within these clouds it is possible to infer the cosmic background temperature at high redshift. Ge *et al* (1997) derive an excitation temperature of approximately 11.6 K for neutral carbon at a redshift of $z = 1.9731$, measured in absorption against a background quasar. This represents an upper limit to the cosmic background temperature at that redshift and by estimating other excitations they find the CBR temperature to be $T_\mathrm{r} = 7.9 \pm 1.0$ K at that redshift, which is consistent with cosmological predictions.

When quasars appeared in the Universe they are believed to have reionized it. There has long been a search for quasars which show a break at the hydrogen ionization energy suggesting that there is some intervening neutral material: the 'Gunn–Peterson test' (1965). Recently, the digital Sloan Survey has found quasars with absorption in the UV suggesting that the Universe was reionized at about $z = 6$ (Becker *et al* 2001).

Since protons have no emission lines, they are not observed directly, but their presence can be inferred if some heavier elements are present. Tripp *et al* (2000) looked at O VI lines and found many of them in low-redshift absorption systems. The presence of highly ionized oxygen between galaxies suggests that there is an extremely hot component to the intergalactic medium.

Quasar emission lines can also be used to test whether physical constants are changing with time. By observing the ratio of several lines from the same cloud it is possible to infer the fine structure constant α; this was first attempted by Savedoff (1956). Early attempts at measuring the fine structure constant used the *alkaline doublet* method wherein measurements are made of doublet absorption lines for which the spacing is proportional to α^2 and so any

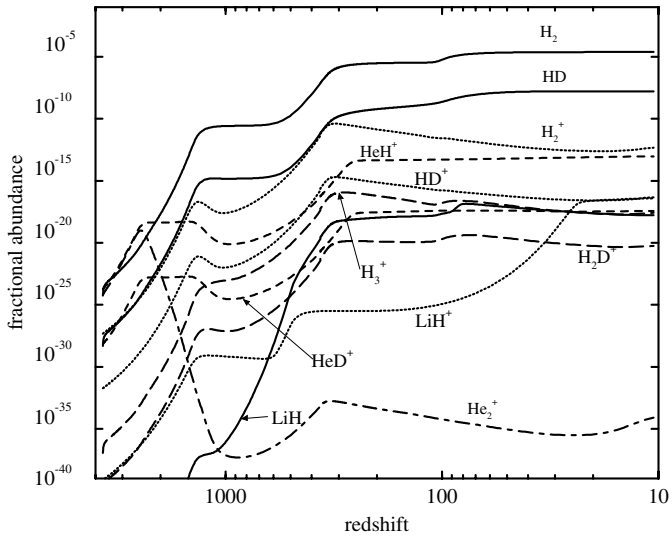


Figure 2. Fractional abundances of molecules and molecular ions in the early Universe at high redshift z. The cosmological parameters correspond to the standard cosmology model III of Stancil *et al* (1996, 1998): $\Omega_0 = 1$, $\Omega_b = 0.0367$, $h = 0.67$.

variation in spacing is proportional to α. The measurements were all consistent with no time dependence and the best limit was $\delta\alpha/\alpha = 3.5\ (\pm 5.5) \times 10^{-6}$ (Cowie and Songaila 1995) back to $z = 2.811$. More recently a new technique, the *many-multiplet* method, has been used which simultaneously compares the wavelengths of many multiplets to get a more sensitive determination (Webb *et al* 1999). The *alkaline doublet* method compares two lines with the same ground state, whereas the *many-multiplet* method compares many lines with different ground states. Using this method, Webb *et al* (2001) suggest that the fine structure constant is varying with time, with $\delta\alpha/\alpha = -0.72\ (\pm 0.18) \times 10^{-5}$ over the redshift range $0.5 < z < 3.5$. The result is provocative, but the analysis to get it is complicated and errors which could result in such a small effect are many including: unresolved velocity structures, uncertainties in wavelength calibrations or laboratory wavelengths, and, since their lines come from different levels, sensitivity to variations in temperature or excitation within the cloud.

4. Standard chemistry in the early Universe

The chemistry of the early Universe was investigated by Lepp and Shull (1984), Dalgarno and Lepp (1987), Latter and Black (1991), Puy *et al* (1993), and Palla *et al* (1995). All of these examined one or more of the molecules which might be formed in the early Universe. More recently, Galli and Palla (1998) and Stancil *et al* (1996, 1998) attempted a comprehensive review of all the reactions involved in the study of the early Universe. In particular, Stancil *et al* (1998) included 23 atomic and molecular species and 144 chemical processes. Figure 2 is adapted from Stancil *et al* (1998) and shows the abundance of molecular ions and molecules with redshift.

In table 1, we list 41 reactions which we have considered since Stancil *et al* (1998). They can be added to the lists given in table 1 of Stancil *et al* (1996, 1998) for a more complete set. Most of these reactions are new to our models, but some are improved rate coefficient fits which supersede those previously given. We will discuss some of these reactions below.

Table 1. Gas-phase reactions and their rate coefficients. The rate coefficient fits are given by the relation $\alpha = a_1(T/300)^{a_2}\exp(-T/a_3)$.

	Reaction	a_1[a]	a_2	a_3 (K)	Notes[d]
(1)	$H + e^- \rightarrow H^+ + 2e^-$	$7.57 - 10$[b]	$5.50 - 1$	$1.582 + 5$[c]	1
(2)	$D + e^- \rightarrow D^+ + 2e^-$	$7.57 - 10$	$5.50 - 1$	$1.582 + 5$[c]	2
(3)	$He + e^- \rightarrow He^+ + 2e^-$	$4.12 - 10$	0.5	$2.85 + 5$[c]	3
(4)	$He^+ + e^- \rightarrow He^{2+} + 2e^-$	$9.84 - 11$	0.5	$6.32 + 5$[c]	4
(5)	$Li + e^- \rightarrow Li^+ + 2e^-$	$3.11 - 8$	$1.63 - 1$	$6.27 + 4^3$	4
(6)	$Li^+ + e^- \rightarrow Li^{2+} + 2e^-$	$5.67 - 12$	$7.15 - 1$	$8.77 + 5$[c]	4
(7)	$Li^{2+} + e^- \rightarrow Li^{3+} + 2e^-$	$1.7 - 12$	$7.09 - 1$	$1.42 + 6$[c]	4
(8)	$He^+ + e^- \rightarrow He + \nu$	$3.66 - 7$	-1.5	$4.7 + 5$[c]	5
(9)	$He^{2+} + e^- \rightarrow He^+ + \nu$	$1.97 - 11$	$-6.34 - 1$	$1.32 + 6$	6
(10)	$Li^+ + e^- \rightarrow Li + \nu$	$1.33 - 8$	-1.5	$6.53 + 5$[c]	7
(11)	$Li^{2+} + e^- \rightarrow Li^+ + \nu$	$1.87 - 11$	$-6.41 - 1$	$1.76 + 6$	6
(12)	$Li^{2+} + e^- \rightarrow Li^+ + \nu$	$5.34 - 8$	-1.23	$9.23 + 5^3$	7
(13)	$Li^{3+} + e^- \rightarrow Li^{2+} + \nu$	$4.83 - 11$	$-6.21 - 1$	$1.67 + 6$	6
(14)	$H^+ + D \rightarrow H + D^+$	$1.97 - 9$	$-4.02 - 1$	$3.71 + 1$	8
		$-1.63 - 13$	1.49	—	
(15)	$D^+ + H \rightarrow D + H^+$	$1.97 - 9$	$-3.95 - 1$	$3.30 + 1$	8
		$3.04 - 10$	$-3.33 - 1$	—	
(16)	$H^+ + H^- \rightarrow H + H$	$1.40 - 7$	$-4.87 - 1$	$-2.93 + 4$	9
(17)	$H^+ + D^- \rightarrow H + D$	$1.61 - 7$	$-4.87 - 1$	$-2.93 + 4$	10
(18)	$H^- + D^+ \rightarrow H + D$	$1.61 - 7$	$-4.87 - 1$	$-2.93 + 4$	10
(19)	$D^+ + D^- \rightarrow D + D$	$1.96 - 7$	$-4.87 - 1$	$-2.93 + 4$	10
(20)	$Li^+ + H^- \rightarrow Li + H$	$2.93 - 7$	$-4.77 - 1$	$-2.32 + 4$	9
(21)	$Li^- + H^+ \rightarrow Li + H$	$1.80 - 7$	$-4.77 - 1$	$-2.32 + 4$	7
(22)	$Li^+ + D^- \rightarrow Li + H$	$3.71 - 7$	$-5.10 - 1$	$-4.41 + 4$	11
(23)	$Li^- + D^+ \rightarrow Li + D$	$2.28 - 7$	$-5.10 - 1$	$-4.41 + 4$	12
(24)	$H + \nu \rightarrow H^+ + e^-$	$1.14 + 7$	0.98	$1.57 + 5$[c]	13
(25)	$D + \nu \rightarrow D^+ + e^-$	$1.14 + 7$	0.98	$1.57 + 5$[c]	2
(26)	$He + \nu \rightarrow He^+ + e^-$	$1.11 + 7$	1.23	$2.80 + 5$[c]	14
(27)	$He^+ + \nu \rightarrow He^{2+} + e^-$	$5.45 + 5$	1.63	$5.90 + 5$[c]	15
(28)	$Li + \nu \rightarrow Li^+ + e^-$	$5.08 + 6$	1.45	$6.05 + 4$[c]	14
(29)	$Li^+ + \nu \rightarrow Li^{2+} + e^-$	$9.23 + 6$	1.37	$8.46 + 5$[c]	15
(30)	$Li^{2+} + \nu \rightarrow Li^{3+} + e^-$	$5.96 + 2$	2.63	$1.27 + 6$[c]	13
(31)	$H^- + \nu \rightarrow H + e^-$	$2.08 + 4$	2.13	$8.82 + 3$[c]	14
(32)	$D^- + \nu \rightarrow D + e^-$	$2.08 + 4$	2.13	$8.82 + 3$[c]	2
(33)	$Li^- + \nu \rightarrow Li + e^-$	$5.29 + 5$	1.40	$8.10 + 3$[c]	14
(34)	$H_2 + D^+ \rightarrow HD + H^+$	$1.60 - 9$	—	—	16
(35)	$H_2^+ + H^- \rightarrow H_3^+ + e^-$	$2.70 - 10$	$-4.85 - 1$	$-3.12 + 4$	17

The chemistry of the early Universe is restricted to gas-phase reactions among only three elements. In the standard Big Bang model, the Universe at recombination had only ^{1}H, ^{2}H (or D), ^{3}He, ^{4}He and ^{7}Li. The radiation was primarily black-body, but a low-intensity diffuse component also resulted from recombination of atoms.

The first molecules and molecular ions must have been formed in the absence of any surfaces or dust grains, which were only created later after the first stars appeared. In the low-density environment after recombination, three-body reactions were inefficient. This leaves only radiative processes to form the first molecules or molecular ions. As soon as neutral atoms appeared, the first molecules were formed through radiative association. The first molecular ions are discussed below, they are helium compounds and were formed when the helium recombined.

Table 1. (continued.)

	Reaction	a_1[a]	a_2	a_3 (K)	Notes[d]
(36)	$H_2^+ + D^- \rightarrow H_2D^+ + e^-$	$2.24-10$	$-4.90-1$	$-4.36+4$	17
(37)	$He_2^+ + e^- \rightarrow He + He$	$6.10-11$	$-9.00-1$	—	18
(38)	$LiH + H \rightarrow Li + H_2$	$2.52-10$	1.0	$2.50+3$	19
		$2.15-10$	0.28	—	
(39)	$LiH + H^+ \rightarrow Li^+ + H_2$	$2.00-15$	—	—	20
(40)	$LiH + H^+ \rightarrow LiH^+ + H$	$2.00-15$	—	—	20
(41)	$LiH^+ + e^- \rightarrow Li + H$	$3.80-7$	-0.47	—	21

[a] For collision processes a_1 is in $cm^3\ s^{-1}$ while for photo-processes it has units of s^{-1}.
[b] The notation $A+B$ corresponds to $A \times 10^B$.
[c] For the reactions indicated the exponential term in the rate coefficient relation has the form $\exp(-a_3/T)$.
[d] Notes: 1, fit to Scholz and Walters (1991); 2, assumed the same as the corresponding H reaction; 3, Black (1981); 4, fit to Voronov (1997); 5, Black (1981), dielectronic recombination (DR); 6, fit to Verner and Ferland (1996), radiative recombination (RR); 7, fit to Badnell (1998), DR; 8, Savin (2002), total rate coefficient is the sum of the two lines; 9, fit to Croft *et al* (1999b); 10, same as corresponding H reaction, but scaled by D reduced mass; 11, determined from cross section of Croft *et al* (1999a); 12, same as (22), but magnitude scaled by (21)/(20); 13, determined from cross section of Osterbrock (1989); 14, Galli and Palla (1998); 15, determined from cross section of Verner *et al* (1996); 16, Wang and Stancil (2002); 17, determined from cross section of Naji *et al* (1998); 18, Carota *et al* (1999); 19, Dunne *et al* (2001), with the former rate coefficient for $T < 280$ K and the latter for $T > 280$ K; 20, Bodo *et al* (2001b); 21, Krohn *et al* (2001).

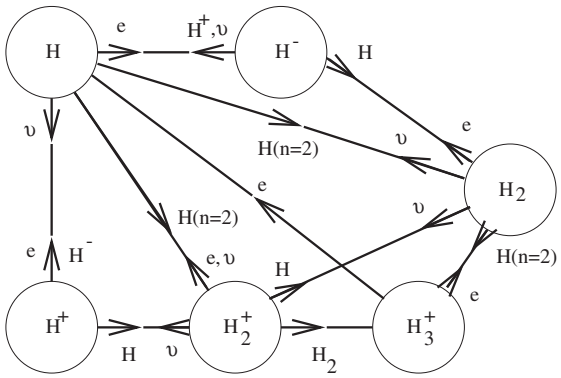


Figure 3. Bubble diagram of hydrogen reactions in the early Universe.

4.1. Hydrogen chemistry

Molecular hydrogen was the first neutral molecule formed in the Universe and remains the most abundant. Because it does not have a dipole moment, molecular hydrogen cannot form directly by a radiative process. The most common reaction paths leading to H_2 formation in the early Universe use H_2^+ and H^- as intermediaries. A bubble diagram showing the hydrogen reactions in the early Universe is shown in figure 3.

The molecular ion HeH^+ was the first to appear (see 'Helium chemistry' below) and so at the earliest times it is the first to produce H_2. The HeH^+ is first converted to the H_2^+ molecular ion and then to H_2 through the series of reactions

$$H^+ + He \rightarrow HeH^+ + \nu, \tag{5}$$

$$HeH^+ + H \rightarrow H_2^+ + He, \tag{6}$$

$$H_2^+ + H \rightarrow H_2 + H^+, \tag{7}$$

where both protons and He acted as catalyst, being returned to the gas once H_2 is formed to start the process again. The H_2 formed this way was quickly destroyed by the CBR and at later times, when the Universe had cooled sufficiently, the H_2^+ was primarily formed directly by the radiative association reaction

$$H^+ + H \rightarrow {H_2}^+ + \nu \tag{8}$$

which then went on to make H_2 by reaction (7) (the so-called H_2^+ channel).

Other reactions which have been proposed to produce H_2 are formation through excited states directly to H_2:

$$H^* + H \rightarrow H_2 + \nu. \tag{9}$$

(Latter and Black 1991) or to the molecular ion:

$$H^* + H \rightarrow {H_2}^+ + e^-. \tag{10}$$

The excited atom H^* can be created either through photoexcitation, electron-impact excitation, or following a recombination. Neither reaction path significantly affects the molecular hydrogen abundance in the early Universe.

Excited atoms may react with molecular hydrogen to produce H_3^+:

$$H^* + H_2 \rightarrow {H_3}^+ + e^-. \tag{11}$$

H_3^+ may also be produced through radiative association:

$$H^+ + H_2 \rightarrow {H_3}^+ + \nu, \tag{12}$$

though the rate coefficient is uncertain (see Stancil *et al* 1998), and by associative ionization:

$$H_2^+ + H^- \rightarrow H_3^+ + e^-. \tag{13}$$

For the latter the cross section was measured by Naji *et al* (1998). As the abundances of both reactants are small, the process has little effect on the reactants or the products, which was explicitly checked by rerunning the chemistry model of Stancil *et al* (1998). Our fit to the rate coefficient determined from the experimental cross section of reaction (13) is given in table 1.

As the radiation temperature cools, molecular hydrogen is formed through H^- by the reactions of radiative attachment:

$$H + e^- \rightarrow H^- + \nu \tag{14}$$

and associative detachment:

$$H^- + H \rightarrow H_2 + e^- \tag{15}$$

(the H^- channel). This sequence is impeded at early times by the rapid photodetachment of the H^- by the background radiation field. However, for $z < 100$ it produces most of the molecular hydrogen as can be seen in figure 2.

Stancil and Dalgarno (1998) explored the possibility that the formation of H^- could be enhanced through the stimulated process

$$H + e^- + \nu_{CBR} \rightarrow H^- + \nu + \nu_{CBR}, \tag{16}$$

where ν_{CBR} is a CBR photon. However, the enhancement is small for $T_r < 3000$ K, though it increases significantly for $T_r > 10\,000$ K.

4.2. Deuterium chemistry

Deuterium is important in the early Universe because deuterated-hydrogen molecules have (instantaneous) dipole moments where the pure hydrogen molecules do not. This is potentially significant in the cooling of gas, where cooling through HD dipole radiation would be effective to a much lower temperature than cooling by quadrupole radiation from H_2. The dipole moments of deuterated molecules are also possibly significant in altering the CBR through Thompson scattering. H_2D^+ has a large dipole moment of approximately 0.6 D and Dubrovich (1993) has proposed that if it was in sufficient abundance it could wash out fluctuations in the CBR by scattering. Detailed calculations (Stancil *et al* 1998) suggest that the abundance of H_2D^+ is too low to significantly scatter background photons.

Deuterium chemistry of the early Universe has been studied by many authors (Lepp and Shull 1984, Dalgarno and Lepp 1987, Latter 1989, Palla *et al* 1995, Stancil *et al* 1998). In general, reactions which proceed for hydrogen atoms will proceed for deuterium as well. Fractionation (when the abundance ratios of deuterated species to hydrogen species are different to the D/H elemental abundance ratio) occurs for two reasons. First, there are some additional channels which are not allowed with pure hydrogen. For example, HD may form directly by radiative association:

$$H + D \rightarrow HD + \nu, \tag{17}$$

which is forbidden between two hydrogen atoms because they are identical particles. Only a small amount of HD is made through this reaction even if stimulated effects due to the background radiation field are considered (Stancil and Dalgarno 1997b).

Additional fractionation occurs because deuterated molecules are often more tightly bound than their pure hydrogen isomers. As a result the reaction of swapping a deuterium atom for a hydrogen atom may proceed faster forward then backward. This results in a significant enhancement in the deuterated fraction mostly through

$$H^+ + D \rightarrow H + D^+ \tag{18}$$

followed by

$$H_2 + D^+ \rightarrow HD + H^+ \tag{19}$$

which is also the main source of HD in interstellar clouds. Reaction (18) and the reverse process

$$D^+ + H \rightarrow D + H^+ \tag{20}$$

have recently been investigated by Igarashi and Lin (1999), Zhao *et al* (2000), and Esry *et al* (2000). Using the cross sections from these studies, Savin (2002) has presented a set of recommended rate coefficients. Replacing the adopted rate coefficients in Stancil *et al* (1998) for reactions (18) and (20) with Savin's results, we find a slight decrease (<30%) in all deuterated species, except for D itself. More important, however, is the value of the rate coefficient for reaction (19). Wang and Stancil (2002) have recently reviewed the available measurements and calculations and note a variation of more than a factor of two for $T < 10\,000$ K. They however presented preferred rate coefficients based on the merged-beams measurement of Gerlich (1992) which is nearly constant for $T < 1000$ K with a value of 1.6×10^{-9} cm^3 s^{-1}. This is about 25% smaller than the value adopted by Stancil *et al* (1998) resulting in a proportional decrease in the HD fractional abundance. As in the hydrogen chemistry, we added the recently measured associative ionization reaction

$$H_2^+ + D^- \rightarrow H_2D^+ + e^-. \tag{21}$$

(Naji *et al* 1998), but found it to have a negligible effect.

4.3. Helium chemistry

The chemistry of helium was investigated by Lepp and Shull (1984) who considered the formation of HeH^+ and recently by Galli and Palla (1998) and Stancil *et al* (1998); the latter also considered the trace molecular ions HeD^+ and He_2^+. Helium has the distinction of being the first neutral atom to recombine and then the first atom to form a molecular bond. The first molecular bond was probably formed in the radiative association reaction

$$He + He^+ \rightarrow He_2^+ + \nu \tag{22}$$

(or reaction (5)), though the temperature of the radiation was still too high for these ions to survive long and He_2^+ was quickly removed by photodissociation:

$$He_2^+ + \nu \rightarrow He^+ + He, \tag{23}$$

as well as by dissociative recombination:

$$He_2^+ + e^- \rightarrow He + He. \tag{24}$$

Also formed and destroyed at this time by similar processes are HeH^+ and $LiHe^+$. Eventually, the helium recombines to near completion and the He_2^+ abundance diminishes with the diminishing He^+ abundance.

Standard helium recombination calculations suggest that He fully recombined with a residual fractional ionization of only $\sim 10^{-23}$, so He_2^+ was produced in substantial abundances only at the epoch of helium recombination, but became negligible thereafter. However, the recent detailed analysis of the early Universe recombination by Seager *et al* (1999, 2000) found that He recombination was delayed somewhat until the time of hydrogen recombination. A competition for available electrons hinders helium recombination so the residual He ionization was increased to $\sim 10^{-6}$. Consequently, the He_2^+ abundance is increased to $\sim 10^{-22}$; still too small to be of any real importance.

HeH^+ was formed at early times by the radiative association processes

$$He^+ + H \rightarrow HeH^+ + \nu. \tag{25}$$

(Roberge and Dalgarno 1982, Kraemer *et al* 1995). After helium began to recombine, HeH^+ formed even earlier through

$$He + H^+ \rightarrow HeH^+ + \nu \tag{26}$$

(Zygelman *et al* 1998). Additional HeH^+ could be made by stimulated radiative association with photons from the CBR. Zygelman *et al* (1998) found that this increased the abundances by about 10%.

Once HeH^+ formed, it contributed to the production of H_2^+ through the exchange reaction (6). A cross section for the exoergic reaction (6) was constructed by Linder *et al* (1995) from experimental data and asymptotic energy extrapolations.

Apparently no information is available concerning HeD^+. In view of this, Stancil *et al* (1998) assumed that the reaction rate coefficients could be estimated by mass-scaling the analogous HeH^+ reactions. As neither HeD^+ nor He_2^+ are thought to have significant influence in the early Universe, investigations of their formation and destruction processes are not an immediate need.

4.4. Lithium chemistry

The chemistry of lithium has received considerable attention recently because of the possible role of LiH in scattering CBR photons in the recombination era (see for example Maoli *et al* (1994)) and as a possible coolant in the gravitational collapse of the first cosmological objects.

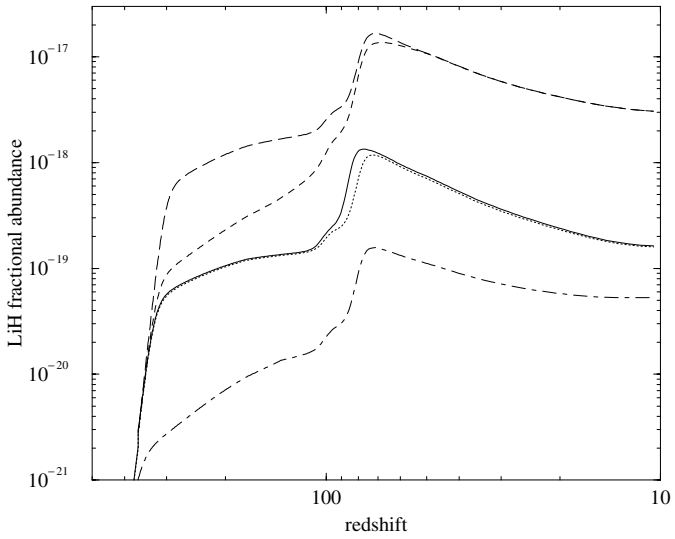


Figure 4. Fractional abundance of LiH in the early Universe. Previous result, Stancil *et al* (1998) (solid curve); utilizing new rate coefficients for mutual neutralization reactions (32), (33) (dotted curve); with a 420 K activation barrier for LiH + H reaction (28) (short dashed curve); with reaction (28) set to zero (long dashed curve); and using the rate coefficients of Dunne *et al* (2001) for reaction (28) (dot–dashed curve).

Recombination era lithium chemistry was first studied by Lepp and Shull (1984) and later by Puy *et al* (1993) and Palla *et al* (1995). In these investigations the major formation mechanism of LiH was taken to be the radiative association process

$$\mathrm{Li} + \mathrm{H} \rightarrow \mathrm{LiH} + \nu. \tag{27}$$

However, reliable rate coefficients for reaction (27) only became available with the quantum mechanical calculations of Dalgarno *et al* (1996) and Gianturco and Gori Giorgi (1997). The new rate coefficients motivated further and more comprehensive investigations of the lithium chemistry by Stancil *et al* (1996), Bougleux and Galli (1997), Galli and Palla (1998), and Stancil *et al* (1998). In particular, the later work resulted in a significant reduction in the predicted LiH fractional abundance from $\sim 10^{-10}$ to $\sim 10^{-19}$, depending on the cosmological model adopted. The new predicted LiH abundance was partially due to a reduction in the rate coefficient for reaction (27) (Dalgarno *et al* 1996, Gianturco and Gori Giorgi 1997), but also as a consequence of the H exchange reaction

$$\mathrm{LiH} + \mathrm{H} \rightarrow \mathrm{Li} + \mathrm{H_2} \tag{28}$$

(see figure 12 of Stancil *et al* (1998)) for which Stancil *et al* (1996) adopted the temperature-independent rate coefficient of 2×10^{-11} cm^3 s^{-1}, assuming the absence of an activation barrier. Subsequently, a spin-coupled valence-bond study of the relevant potential surfaces for reaction (28) found a small barrier of 0.036 eV ($E/k = 420$ K) in the collinear arrangement (Clarke *et al* 1998). While figure 12 of Stancil *et al* (1998) shows that reaction (28) is the primary LiH destruction mechanism for $z < 350$ ($T_\mathrm{m} < 800$ K), the threshold predicted by Clarke *et al* (1998) suggests that the destruction efficiency would be reduced. We demonstrate in figure 4 that the 420 K threshold allows for an order of magnitude increase of the LiH fractional abundance. However, Dunne *et al* (2001) recently calculated a complete potential energy surface for LiH_2 and found no energy barrier. After performing quasi-classical scattering calculations on this surface, they find a rate coefficient

of $8.4 \times 10^{-13} T e^{-0.0004T}$ cm^3 s^{-1} between 280 and 4000 K and $4.36 \times 10^{-11} T^{0.28}$ cm^3 s^{-1} for $T < 280$ K. This results in an order of magnitude reduction in the LiH abundance suggesting that the uncertainty in the fractional abundance of LiH could be as large as two orders of magnitude. LiH_2 potential surfaces were also computed by Lee *et al* (1999) who failed to detect a barrier, and the collinear work of Clarke *et al* (1998) was extended to other orientations by Bodo *et al* (2001a). Bodo *et al* (2001a) found that the barrier was only present for orientation angles less than $\sim 20^\circ$. The evidence that reaction (28) is not hindered by an activation barrier is accumulating. Further, Lee *et al* (1999) proposed that reaction paths to form Li(2p):

$$LiH + H \rightarrow Li(2p) + H_2 \tag{29}$$

might also be very efficient. Therefore, the suggestion by Stancil *et al* (1996) that the LiH destruction process (28) should proceed with a relatively fast rate appears to be correct. However, experiments and further calculations are needed to confirm the magnitude of the rate coefficients predicted by Dunne *et al* (2001).

Stancil *et al* (1996) pointed out that LiH could be formed by the associative detachment processes

$$Li^- + H \rightarrow LiH + e^- \tag{30}$$

and

$$Li + H^- \rightarrow LiH + e^-. \tag{31}$$

With the estimate of 4×10^{-10} cm^3 s^{-1} for the rate coefficient for both reactions (30) and (31), it was found that for $z < 100$, reaction (30) was the dominant LiH production mechanism, with reaction (31) giving some contribution. The LiH fractional abundance remains uncertain as no experimental or theoretical analysis is available for either reaction.

Production of LiH via either reactions (30) or (31) requires a significant abundance of neutral Li which is primarily formed through radiative recombination. However, some neutral Li can be made by the mutual neutralization process

$$Li^+ + H^- \rightarrow Li + H. \tag{32}$$

Stancil *et al* (1996, 1998) adopted the theoretical rate coefficients of Janev and Radulović (1978), but reduced by about a factor of two to match the $Li^+ + D^-$ merged-beams measurement of Peart and Hayton (1994). Recently, an extensive *ab initio* molecular-orbital close-coupling analysis by Croft *et al* (1999a) revealed that the measurement underestimated the true magnitude of the cross section due to the limited acceptance angle of the detector. Additionally, they demonstrated that for low collision energies, the so-called focusing effect (or kinematic isotope effect—see for example Stancil and Zygelman (1995))—enhances the $Li^+ + H^-$ cross sections compared to that of $Li^+ + D^-$. As a consequence they computed new rate coefficients based on their theoretical cross section (Croft *et al* 1999b). However, as the theoretical cross section for $Li^+ + D^-$ is $\sim 20\%$ larger than the measured cross section corrected for the finite acceptance angle (Croft *et al* 1998), we recommend the rate coefficients be reduced by 20%. A fit to the rate coefficient with the correction is given in table 1. Incidentally, the recommended rate coefficients are very similar to the Landau–Zener results of Janev and Radulović (1978) with the discrepancy in the rate coefficients adopted by Stancil *et al* (1996) being a consequence of the finite acceptance angle and focusing effect. The reaction turns out to have only a small influence on the Li^+/Li abundance ratio.

The analogous mutual neutralization process

$$Li^- + H^+ \rightarrow Li + H \tag{33}$$

is of considerably more importance, as it can deplete the Li^- abundance, hindering LiH formation through reaction (30). Using the Landau–Zener approximation, Croft *et al* (1999b)

estimated that the rate coefficients should be only slightly larger than those adopted by Stancil *et al* (1996). The agreement is rather fortuitous as Stancil *et al* (1996) made the arbitrary assumption that reactions (32) and (33) would proceed with the same probability, when in fact Croft *et al* (1999b) find the rate coefficient ratio (32)/(33) to be ∼1.6. We list a fit to the new rate coefficients in table 1 and show in figure 4 that it changes the LiH abundance only slightly.

The possibility that reactions which are stabilized by the emission of radiation may be enhanced by the background radiation field was mentioned above for H^-, HD, and HeH^+. This effect was also considered for radiative attachment to form Li^- (Stancil and Dalgarno 1998) and for radiative association to form LiH (Stancil and Dalgarno 1997a). As in the case of H^-, stimulated radiative attachment of Li^- gives a small enhancement for $T_r < 3000$ K while the abundance of LiH in the early Universe was found to only be increased by ∼25%.

Finally, for LiH, Stancil *et al* (1996), in their elaborate chemistry, introduced the following LiH destruction processes due to proton impact:

$$LiH + H^+ \rightarrow Li^+ + H_2, \tag{34}$$

$$LiH + H^+ \rightarrow Li + H_2^+, \tag{35}$$

and the charge transfer reaction

$$LiH + H^+ \rightarrow LiH^+ + H. \tag{36}$$

As all three processes are exoergic, they assumed a large temperature-independent rate coefficient of 10^{-9} cm^3 s^{-1} for each. Recently, Bodo *et al* (2001b) have studied the potential energy surfaces of all possible reactions for the LiH_2^+ system including processes (34)–(36). They argued that because of the large exoergicities of reactions (34) and (36), 5.8 and 10.2 eV, respectively, and the apparent lack of avoided crossings, the rate coefficients would be small with an upper limit of $\sim 2 \times 10^{-15}$ cm^{3}s^{-1}. On the other hand, the small 0.13 eV exoergicity of reaction (35) and the lack of an activation barrier led Bodo *et al* (2001b) to conclude, in agreement with Stancil *et al* (1996), that the process would be efficient. However, because of the relatively small proton abundance (in comparison to H) after recombination, all three processes turn out to be, at most, only secondary destruction mechanisms with rates about two orders of magnitude smaller than the dominant process (28) as shown in figure 12 of Stancil *et al* (1998), if the large rate coefficients of Stancil *et al* (1996) are adopted. Therefore these processes, with the possible exception of (35), have no influence on the abundances of any of the reactants or products. The uncertainty in the LiH abundance is therefore governed by the limited knowledge of processes (28)–(31) for which measurements and calculations are needed.

The lithium hydride ion LiH^+ was first introduced into the study of early Universe chemistry by Dalgarno and Lepp (1987) where it is formed by the ion–atom radiative association processes

$$Li^+ + H \rightarrow LiH^+ + \nu \tag{37}$$

and

$$Li + H^+ \rightarrow LiH^+ + \nu. \tag{38}$$

Interestingly, they suggested that it might be more abundant than LiH as a consequence of a large residual atomic lithium ionization fraction. Once accurate radiative association rate coefficients became available (Dalgarno *et al* 1996, Gianturco and Gori Giorgi 1997), this prediction was demonstrated in chemical models (Bougleux and Galli 1997, Galli and Palla 1998, Stancil *et al* 1998). For $z < 25$, LiH^+ was found to be more abundant than LiH by an order of magnitude.

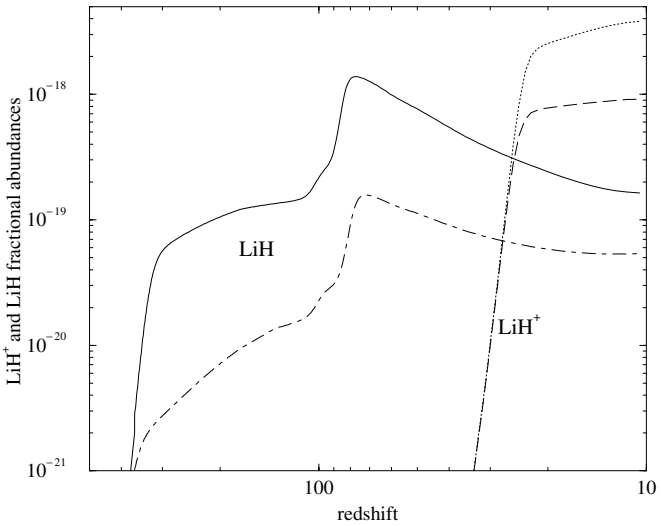


Figure 5. Fractional abundance of LiH^+ in the early Universe. Previous result, Stancil *et al* (1998) (dotted curve), and utilizing new rate coefficients for the dissociative recombination reaction (39) (dashed curve). Comparison is made to the LiH abundance calculations from figure 4: Stancil *et al* (1998) (solid curve), and using the rate coefficients of Dunne *et al* (2001) for reaction (28) (dot–dashed curve).

An elaborate LiH^+ chemistry was first described by Stancil *et al* (1996)[4], though the majority of the reaction rate coefficients were, and still are, unknown. While LiH^+ is primarily formed by radiative association and destroyed by the reverse processes of reactions (37) and (38), photodissociation (see figure 11 of Stancil *et al* (1998)), an important removal mechanism is the dissociative recombination reaction

$$LiH^+ + e^- \rightarrow Li + H. \tag{39}$$

Due to the lack of any data for the reaction, Stancil *et al* (1996) adopted the theoretical rate coefficient for the analogous process on HeH^+. However, a recent storage-ring measurement has found the rate coefficient for reaction (39) at room temperature to be an order of magnitude larger than the HeH^+ value (Krohn *et al* 2001). This results in a reduction in the LiH^+ abundance by a factor of $\sim$3 as shown in figure 5 for $z < 25$. The reduction was not larger because another process within the LiH_2^+ system,

$$LiH^+ + H \rightarrow Li^+ + H_2, \tag{40}$$

was also found to be an important destruction mechanism by Stancil *et al* (1996, 1998). Stancil *et al* (1996) also included

$$LiH^+ + H \rightarrow Li + H_2^+ \tag{41}$$

and the reverse of the charge transfer reaction (36), but the latter two are negligible as they are endoergic by 5.7 and 5.8 eV, respectively. The forward charge transfer reaction (36), even assuming the large rate coefficient adopted by Stancil *et al* (1996), was found to be only a minor LiH^+ production route, being about a factor $\sim 10^4$ less efficient than (38). Bodo *et al* (2001b) argued that instead of (40), which is exoergic by 4.3 eV, the dissociative channel, i.e. $Li^+ + H + H$, would be more probable as some of the product kinetic energy would be transferred

[4] Due to an error in the rate coefficient fit for Li^- radiative attachment, the abundance of LiH was overestimated in Stancil *et al* (1996) so it was always larger than that of LiH^+ (see Bougleux and Galli (1997) and Stancil and Dalgarno (1997a)).

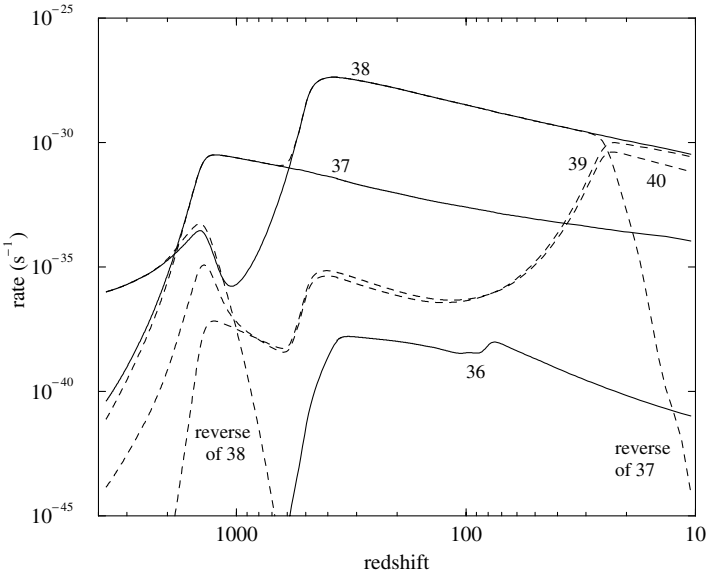


Figure 6. Rates of dominant LiH^+ formation (solid curves) and destruction (dashed curves) processes for model III of Stancil *et al* (1998). Numbers refer to reaction numbers in the text.

to dissociating H_2. However, this is irrelevant from the point of view of LiH^+ and the reaction has no effect on the product abundances. Using the smaller rate coefficient for (36), its LiH^+ production is reduced, but it was already of little importance. To illustrate the effect of the new adopted rate coefficients we show in figure 6 the rates for the dominant LiH^+ formation and destruction processes (we neglect any deuterated species here). This should replace figure 11 of Stancil *et al* (1998) for which we note that the reverse of reaction (37), photodissociation, was left out. From figure 6 it is clear that the uncertainty in the LiH^+ abundance is dependent only on the lack of an accurate rate coefficient for reaction (40) or its dissociative channel. Finally, it appears that regardless of the adopted rate coefficients, LiH^+ remains more abundant than LiH for small redshifts as indicated by figure 5.

5. Rotational–vibrational state-resolved chemistry

To calculate the total abundances (or number density n_x) of each species x, one commonly solves time-dependent chemical rate equations of the form

$$\frac{\mathrm{d}n_x}{\mathrm{d}t} = \alpha_x^{\mathrm{form}}(T_{\mathrm{m}})n_y n_w - \alpha_x^{\mathrm{dest}}(T_{\mathrm{m}})n_x n_u - \zeta_x^{\mathrm{dest}}(T_{\mathrm{r}})n_x + \zeta_x^{\mathrm{form}}(T_{\mathrm{r}})n_z + \cdots \tag{42}$$

where α_x^{form} and α_x^{dest} are rate coefficients for two-body formation and destruction and ζ_x^{form} and ζ_x^{dest} are rates for photo-formation and photo-destruction which depend on the matter temperature T_{m} and the radiation field or radiation temperature T_{r}. This results in a set of coupled stiff differential equations, one for each species of atom, ion, or molecule. In most astrochemical modelling, it is assumed that the rotational–vibrational (RV) distribution of the molecules is in local thermodynamic equilibrium (LTE) or that the entire population resides in the ground RV level. This is generally a reasonable assumption and, in fact, a necessity as little information is available on RV-state-to-state reaction rates.

In situations where it is desirable to determine the RV-level populations, i.e. to model the line emission, time-dependent level-population rate equations must be solved:

$$\frac{\mathrm{d}n_{x,i}}{\mathrm{d}t} = \sum_{j\neq i} n_{x,j}[A_{x,ji} + B_{x,ji}J_{ji}(T_\mathrm{r}) + C_{x,ji}(T_\mathrm{m})] - n_{x,i}\sum_{j\neq i}[A_{x,ij} + B_{x,ij}J_{ij}(T_\mathrm{r}) + C_{x,ij}(T_\mathrm{m})], \tag{43}$$

where the total number density of species x is

$$n_x = \sum_j n_{x,j}. \tag{44}$$

$A_{x,ij}$ and $B_{x,ij}$ are the spontaneous and stimulated radiative transition probabilities and $C_{x,ij}$ are the collision-induced transition probabilities for species x between levels i and j. J_{ij} is the radiation field intensity.

However, it is beginning to be recognized that, in many environments, if accurate abundances and physical parameters are to be predicted or derived from observations of astrophysical objects, in addition to the necessity of following the time-dependent evolution of the particle abundances and level populations, these two cannot be decoupled, but must be solved self-consistently. Therefore, equations (42) and (43) should be replaced by

$$\frac{\mathrm{d}n_{x,i}}{\mathrm{d}t} = \alpha^{\mathrm{form}}_{x,i}(T_\mathrm{m})n_{y,k}n_{w,l} - \alpha^{\mathrm{dest}}_{x,i}(T_\mathrm{m})n_{x,i}n_{u,m} - \zeta^{\mathrm{dest}}_{x,i}(T_\mathrm{r})n_{x,i} + \zeta^{\mathrm{form}}_{x,i}(T_\mathrm{r})n_{z,o} + \sum_{j\neq i} n_{x,j}[A_{x,ji} + B_{x,ji}J_{ji}(T_\mathrm{r}) + C_{x,ji}(T_\mathrm{m})] - n_{x,i}\sum_{j\neq i}[A_{x,ij} + B_{x,ij}J_{ij}(T_\mathrm{r}) + C_{x,ij}(T_\mathrm{m})]. \tag{45}$$

We are aware of some preliminary applications of this approach to interstellar clouds (Black 1998). However, in a related context, Galli and Palla (1998) discussed the influence of RV-specific photodissociation of H_2^+ on the H_2^+ and H_2 abundances in the recombination era. Typically, an LTE photodissociation rate is adopted, but if photodissociation from specific RV levels

$$H_2^+(v, J) + \nu \rightarrow H + H^+, \tag{46}$$

for example from the ground level or some high-lying level, is considered, the abundance of H_2^+ can vary by orders of magnitude. Further, as H_2 is formed primarily by

$$H_2^+(v, J) + H \rightarrow H_2(v', J') + H^+ \tag{47}$$

for $130 < z < 700$, the abundance of H_2 can also be severely affected. This has consequences for the later evolution of the Universe when primordial clouds collapse out of the uniform primordial gas. Heat generated during the collapse is radiated away by H_2, as discussed below. Changes in the initial H_2 abundances can affect the cooling timescale and hence when the first galaxies and stars formed.

As hinted at earlier, the major obstacle to modelling H_2 and H_2^+ with equation (45) is the lack of RV-state-to-state reaction data. Work is in progress to extend the earlier calculations on (46) (Stancil 1994) and its reverse process, the radiative association reaction

$$H^+ + H \rightarrow H_2^+(v, J) + \nu, \tag{48}$$

to obtain RV-state-to-state rate coefficients. Data on the dissociative recombination

$$H_2^+(v, J) + e^- \rightarrow H + H \tag{49}$$

and associative detachment

$$H^- + H \rightarrow H_2(v, J) + e^- \tag{50}$$

have been computed by (equation (49)) Schneider *et al* (1994, 1997) and (equation (50)) Bieniek and Dalgarno (1979) and Launay *et al* (1991). However, the information on reaction (47) is limited to the calculation of the reverse process

$$H^+ + H_2(v, J) \rightarrow H_2^+(v', J') + H \tag{51}$$

by Ichihara *et al* (2000, 2001), and Krstić *et al* (2002) with the lowest energies considered being 0.1, 3, and 0.6 eV, respectively. While the latter two works give vibrationally resolved state-to-state cross sections making it possible to use detailed balance to estimate the rate coefficients for reaction (47), the calculation of Ichihara *et al* (2000) is only initial vibrational-state-resolved, being summed over all final channels. More work, including J-resolution, is needed.

A similar analysis would be useful for HD and HD^+, but the only state-selective data that we are aware of is the vibrationally resolved dissociative recombination,

$$HD^+(v) + e^- \rightarrow H + D, \tag{52}$$

measurements of Amitay *et al* (1998). Further, HD is primarily formed by

$$D^+ + H_2(v, J) \rightarrow HD(v', J') + H^+, \tag{53}$$

for which the available data are mostly limited to total rate coefficients or cross sections as reviewed by Wang and Stancil (2002). However, Ichihara *et al* (2001), using the trajectory surface hopping method, have computed state-to-state cross sections for $v = 0 - 8$ and $J = 1$ to $v' = 0 - 9$, but only for 3–8 eV, while lower energies are required for early Universe studies.

6. Cooling processes

Cooling is important in the early Universe in that it controls the collapse of the first objects (Rees and Ostriker 1977, Silk 1977). The objects must radiate their gravitational energy in order to collapse. It is also possible that this radiation may some day be detectable. Thus, it is important to calculate the cooling correctly, both to get the scale of the first objects and to make predictions of which lines would be observable.

6.1. Atomic cooling

Atomic cooling is primarily effective at high temperatures and, because hydrogen and helium dominate the cooling, is not much different to that for the high-temperature component of the interstellar medium (Dalgarno and McCray 1972). Hydrogen cooling through Lyman α becomes effective at about 8000 K; at higher temperatures the gas becomes ionized and the cooling is primarily through bremsstrahlung.

The cooling by atomic hydrogen occurs when a level is collisionally excited by electron impact:

$$H + e^- \rightarrow H^* + e^- \tag{54}$$

followed by spontaneous emission of a photon:

$$H^* \rightarrow H + \nu. \tag{55}$$

Since the lowest electronic level of hydrogen is 10.2 eV this will primarily occur at high temperature. The lowest level for helium is at 19.8 eV, requiring even higher temperatures for cloud collapse. The lower energy hyperfine transitions have A-values too small to provide significant cooling. The lowest excited electronic energy level for lithium is at 1.8 eV, but the relative abundance of lithium is so small that it is probably not an effective coolant for cloud collapse, though it appears not to have been considered in a detailed calculation. To get significant cooling below 8000 K one must form molecules.

6.2. Molecular cooling

As clouds collapse, the density increases and the rates for molecule formation outlined above, which go as density squared, again become fast compared with other evolution times. The clouds begin to form more molecules which provide cooling below 8000 K. Additionally, as the densities get much higher, three-body processes (discussed below) dominate the formation and the clouds may become nearly completely molecular.

The most abundant molecule, and the most important for the cooling, is H_2. Molecular hydrogen cooling occurs when a collisional excitation

$$H_2(v, J) + H \rightarrow H_2(v', J') + H \tag{56}$$

is followed by a spontaneous emission, cooling the cloud. Since, H_2 lacks a dipole moment, the emission is quadrupole and so J changes by 2 or 0. To calculate the cooling requires a complete set of these state-to-state collision rates as well as rates to other electronic states if one wishes to calculate dissociative cooling, in which a collision leads to direct dissociation of the H_2:

$$H_2 + H \rightarrow H + H + H. \tag{57}$$

Calculations of these collisional rates are sensitive to the potential surface used (Lepp *et al* 1995). Taking advantage of new collisional rates (Martin *et al* 1996, Flower and Roueff 1998) for state-to-state excitation, there have recently been a number of calculations of the H_2 cooling function (Martin *et al* 1996, Tiné *et al* 1998, Le Bourlot *et al* 1999).

Though H_2 usually dominates the cooling, there are conditions where even the smaller amounts of HD can be significant (Dalgarno and Wright 1972). This is because HD has a small dipole moment and thus bigger transition rates; also it allows transitions in which J changes by one. This means that HD has a lower transition energy available to it, $J = 1 \rightarrow 0$, and may well be important at lower temperatures. Flower *et al* (2000) calculated a cooling function for HD and in some cases it can be important in post-shock gas in the early Universe (Flower and Pineau des Forêts 2001).

Other species are only at trace abundances and so even though LiH has a much bigger dipole moment, it is not in most cases an important coolant. An exception might be late stages of collapse of the first objects as it may survive longer than H_2.

Cooling by molecular ions may be important in some cases, particularly if there is some reionization. Neale *et al* (1996) give a cooling function for H_3^+. A cooling function for HeH^+ has yet to be constructed though there have recently been calculations of electron excitation rates

$$HeH^+(v, J) + e^- \rightarrow HeH^+(v', J') + e^- \tag{58}$$

by Rabadán *et al* (1998a, 1998b).

7. The first cosmological objects

We have seen that at recombination the Universe was highly uniform, whereas today the Universe has condensed into stars, galaxies, and clusters of galaxies. How and when this occurred is one of the outstanding problems in astrophysics. We know that the first objects must have collapsed before $z \approx 5$ as quasars are seen at that time. We also know that the Universe had reionized by this time due to observed absorption spectra of quasars. What we also know is that the process was controlled by atomic and molecular physics.

The relevant times for an object to collapse are: the free-fall time, the time it takes to collapse in free-fall, approximately $(G\rho)^{-1/2}$, where G is the gravitational constant and ρ is

the density; the sound crossing time, the time for a pressure wave to cross the object, R/c_s, where R is the size of the object and c_s the sound speed; and a characteristic cooling time for the object, the time it takes to radiate away its energy, $(3n_{\rm H}kT)/(2\Lambda(T))$, where Λ is a cooling function whose determination is discussed in the previous section. The minimum mass at which a cloud is unstable to collapse is known as the Jeans mass, and though a detailed stability analysis is required to get the exact mass, it occurs approximately when the cloud is big enough that its free-fall collapse time is smaller than the sound crossing time. The Jeans mass, M_J, is given by

$$M_J = 2 \times 10^5 \, M_\odot \left(\frac{T_{\rm m}}{100 \, {\rm K}} \right)^{3/2} n^{-1/2}, \tag{59}$$

where $M_\odot$ is the mass of the Sun, T is the temperature, n is the density. Objects with mass greater than M_J are unstable to gravitational collapse. Peebles and Dicke (1968) proposed that the first objects collapsed beginning shortly after the recombination era when the Jeans mass was approximately $10^5 \, M_\odot$. They further proposed that molecular hydrogen formed during the collapse would provide cooling for the cloud.

Collapsing clouds may only continue to collapse if they can radiate away the gravitational potential energy. If the cooling time is less than the free-fall time, then the object can collapse in a free-fall time. If, instead, the cooling time is greater, then the object will collapse until it heats up to some virial temperature, where the pressure forces balance gravitational forces. It then continues to collapse quasi-statically at the slower cooling rate as gravitational energy is radiated away. As the density increases, the cooling rate can increase as well, both from an increase in density and possibly from an increase in the cooling function. Thus an object may collapse quasi-statically for some time and then make a transition to free-fall collapse when the cooling time gets small enough (Fall and Rees 1985).

Palla *et al* (1983) and Lepp and Shull (1984) performed preliminary calculations of the evolution of a cloud under free-fall collapse. The cloud collapses and the abundance of molecular hydrogen is calculated. Initially the molecular hydrogen is at the primordial value, but as the density increases more molecules can be produced. When the density reaches about 10^9 cm^{-3}, three-body reactions, such as

$$\mathrm{H + H + H \rightarrow H_2 + H}, \tag{60}$$

$$\mathrm{H + H + H_2 \rightarrow H_2 + H_2}, \tag{61}$$

become important and nearly all the gas can become molecular. Palla *et al* (1983) followed a 1000 $M_\odot$ cloud and found that the Jeans mass fell to 0.1 $M_\odot$ suggesting it may be possible for a primordial object to fragment into very low mass stars. A heuristic estimate of the effects of shocks on the same cloud however suggests that the Jeans mass never drops below 45 $M_\odot$. Lepp and Shull (1984) included additional cooling by HD and LiH, but got substantially similar results. Modern calculations of primordial cloud evolution are done with high-resolution three-dimensional hydrodynamic simulations including dark matter dynamics (see for example Abel *et al* (2000)). They typically find that a central core develops from the merger of multiple filamentary-like structures with temperature dropping to as low as $\sim$200 K and a central mass of $\sim$200 $M_\odot$. However, these calculations have to date only included H_2 cooling.

8. Shocks

Shocks must occur during the collapse of gas in the early Universe. Even in a uniform collapsing cloud the centre gets optically thick first and stops collapsing, and as the outer parts of the cloud continue to collapse a shock must form. Shocks may also occur in expanding

voids, in cloud–cloud collisions and also as the result of blast waves driven by emission from already formed objects. A shock compresses and heats the gas. This increase in density and temperature can increase the formation of molecules causing the shocked gas to collapse and cool. A similar mechanism is proposed to drive star formation.

Shocks are also important in scenarios of biased galaxy formation. Ostriker and Cowie (1981) and Ikeuchi (1981) proposed a model where first a single heavy object forms, undergoes a burst of star formation, and injects energy into the local intergalactic medium. This makes an expanding shell similar to a large supernova explosion which sweeps up the intergalactic material and forms a shock. If the shock cools sufficiently, a burst of galaxy formation occurs on the surface of the shell. An expanding shell created by the expansion of a void works just as well.

Molecular cooling is critical in getting such a scheme to work (Vishniac *et al* 1985). The Jeans mass is determined by the temperature. Thus the cooling determines the size of the collapsed objects formed or whether they can form at all. In order to form galaxies it is necessary to cool the shock-compressed material well below the 8000 K that can be reached through atomic cooling alone. In the early Universe in the absence of dust or heavy elements, molecules are the only mechanism for cooling the gas to below 8000 K. Without molecular cooling the shell does not become Jeans unstable and would not collapse to form galaxies, possibly instead forming Lyman α systems.

It is possible that ionization from the central object stimulates additional molecule formation (Haiman *et al* 1996). There is a delicate balance as the ionizing photons will destroy molecules nearby. If the radiation is sufficiently hard, the x-ray flux will raise the ionization fraction over a larger region which stimulates molecular production through the production of more H^-. Silk and Rees (1998) used this effect to propose that a quasar may be a black hole that formed first and then stimulated a galaxy to form around it. Additionally, quasars can affect their surroundings by outflows carrying heavy elements out into the intergalactic medium.

The first models of chemistry in intergalactic shells (Struck-Marcell 1982, Wandel 1985) used equilibrium chemistry and found that post-shock gas may cool by forming H_2. MacLow and Shull (1986) solved for a nonequilibrium chemistry and included additional species, most notably deuterium. Shapiro (1986), Shapiro and Kang (1987), and Kang and Shapiro (1992) produced a set of models with nonequilibrium chemistry and increasingly sophisticated hydrodynamic modelling and radiative transfer. Still, the basic conclusion has remained that the gas can form molecules and cool to approximately 100 K behind the shock. Uehara and Inutsuka (2000) have recently reinvestigated this scenario with an improved deuterium chemistry. Their results suggest that nearly all the deuterium will become locked up in HD and the additional cooling lowers the Jeans mass to of order 0.1 $M_\odot$, possibly leading to the formation of brown dwarfs.

The shocks expected in the early Universe have velocities of order 100 km s^{-1} and can heat the gas to temperatures of order 10^5 K. The gas initially cools by inverse Compton scattering and then, as it recombines, by Lyman α cooling. The time for the gas to cool is shorter than the time to recombine so a high residual ionization fraction is left when the gas is cooled to around 8000 K, below which Lyman α cooling is not effective. At this point, molecules can be formed through both the H_2^+ process and the H^- process described above. Initially, the molecules are quickly destroyed through collisional dissociation; still this cools the gas, as much of the heat of formation was radiated away. After the temperature cools to a few thousand degrees the molecules survive and cool the gas radiatively. At this point it is a competition between molecular formation and recombination. The fraction of H_2 is determined by the rates for formation versus the rates for recombination. For a fully ionized gas recombining, this leads to a fractional abundance of H_2 of approximately 10^{-3}.

A higher fractional abundance of H_2 might be produced if there is some external source of ionization, or ionizing photons from the shocked region penetrate into the molecular region. Too much ionization will prevent the molecules from forming and keep the gas from cooling below 8000 K. Kang and Shapiro (1992) investigated models with both ionization from shocks as well as external ionization due to quasars and found that many of the shocks never cooled or collapsed within the age of the Universe. However, as mentioned above, Haiman *et al* (1996) propose increased H_2 production, and thus increased collapse and galaxy formation, due to quasar reionization. The details of the quasar ionizing flux, as well as the relevant atomic and molecular physics, are critical to our understanding of when galaxy formation is enhanced and when it is inhibited.

9. Summary

We have seen that atomic and molecular processes control the evolution of the Universe after recombination. Atoms and molecules are also important probes into our understanding of the early Universe. There is much work still to be done, both in astrophysical modelling of the first objects and in the atomic and molecular physics needed by these models or to test them. There have been great advances in recent years. Many of the reactions used by our models are based on laboratory measurements or theoretical calculations. Improved calculations of molecular structure and rate coefficients have led to better cooling functions. Laboratory measurements of spectroscopically accurate wavelengths have helped spur the measurements of the time variation of the fine structure constant, as well as other probes of astrophysical environments. Better astronomical observations, particularly from the Keck Telescope, Sloan Survey and Hubble Space Telescope, have probed the early Universe environment and are setting limits on fractional abundances and physical conditions. Satellite and balloon observations along with supernova measurements have started to set strong limits on the mass and energy density of the Universe. New satellites will tighten these limits. It is only by continued advances in both astrophysics and atomic and molecular physics that we can hope to understand how our Universe came to be as we find it today.

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